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Research Paper



Vortex-enhanced jet impingement and the role of impulse generation rate in heat removal using additively manufactured synthetic jet devices

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ABSTRACT

This article presents an approach to the design and fabrication of synthetic jet devices (SJDs) using rapid prototyping via additive manufacturing, marking the first study to employ this method for such devices. This manufacturing technique empowers researchers with complete design freedom, enabling the production of ultra-thin SJDs—as thin as 4 mm—without mechanical fasteners and facilitating the rapid fabrication of multiple devices with varying geometries. To showcase the potential of this method, SJDs with conical and cylindrical cavities and orifices ranging from 1.6 mm to 7 mm were designed, fabricated, and tested.

These devices achieved air jet exit velocities exceeding 106 m/s using a single piezoelectric diaphragm—among the highest reported in the literature—validating the effectiveness of this manufacturing approach. This high jet velocity is significant for practical applications requiring efficient thermal management, such as cooling high-power-density electronics, where compact and energy-efficient solutions are essential. Beyond achieving high velocities, it was revealed that maximizing jet velocity alone is not always optimal for heat removal. The hydrodynamic impulse generation rate was introduced as a more significant factor influencing heat transfer performance. By fabricating and testing multiple SJDs with different geometries, it was demonstrated that the impulse generation rate, which accounts for both jet velocity and flow rate, better correlates with enhanced heat transfer capabilities than jet velocity alone. This insight addresses an often-overlooked parameter in SJD design and has substantial implications for optimizing heat removal performance. Moreover, lumped element modeling, tuned solely on diaphragm deflection behavior, accurately predicted device performance and was validated using a hotwire anemometer. This model effectively characterizes center-axis orifice devices and confirms its applicability to thin-cavity designs, providing a valuable tool for future SJD development. Despite moderate volume flow rates (0.2 to 0.8 m^3/h), the fabricated SJDs delivered significant improvements in heat transfer. Compared to natural convection, these devices achieved over 13 times greater heat removal rates, with an average heat transfer coefficient exceeding 120 $W/m^2{\cdot}K$ over a 30 mm × 30 mm heated surface. These findings demonstrate the practicality and effectiveness of vortex-enhanced synthetic jet impingement for targeted and efficient cooling of localized hot spots. This approach offers multiple advantages over traditional rotary cooling systems like fans, including increased reliability, lower profile, while consuming less than 100 mW. The ability to rapidly prototype and optimize SJDs using additive manufacturing accelerates research and development in this field, paving the way for advanced thermal management solutions in real-world applications.

1. Introduction

Jet impingement is an attractive solution for hot spot cooling, especially in electronics [1]. Kwon et al. [2] shows a heat transfer coefficient up to 17.0 $kW/(m^2{\cdot}K)$ at a heat flux of 58.4 W/cm^2 from a continuous air jet at a Reynolds (Re) number of 8.77×10^4. Such high-velocity jets require a high-pressure source. As a result, the design is not portable and will need three major components: a compressor/pump, storage, and piping/coupling/nozzle(s). Fans, air pumps, piezoelectric fans, and synthetic jet devices (SJDs) can be used to avoid such arrangements while producing jet or jet-like flows in a small and portable package.

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Fans are the most common method of cooling electronics, and they use a small electric motor rotating a shaft with attached blades. The generated angular momentum is transferred to air. Modern fan designs are slim and almost silent. However, they have a number of limitations. In small scales, the blade-to-shaft ratio decreases, resulting in less air movement per rotation, and lower overall efficiency. Smaller motors are also less efficient due to thermal and magnetic losses. Moreover, in fans airflow disperses; Axial fans move air parallel to the shaft, with airflow spreading after exit. Centrifugal fans, which move air perpendicular to the shaft, are better suited for controlled airflow. Hofer et al. [3] analyzed the velocity and structure of a miniature centrifugal fan (65.3 × 64.5 × 23.5) mm^3 using particle image velocimetry (PIV) and hot wire anemometry. An average outlet velocity of 9.4 m/s has been measured for a maximum flow rate of 13.54 m^3/h. They showed two counter-rotating vortices at the fan outlet, resulting in a distinctive elliptical, rotating jet that deviates from perpendicular propagation relative to the outlet plane. Yen and Liu [4] compared two centrifugal fans with the same geometric design (60 × 60 × 11 mm^3) but operating at different rotational speeds under identical voltage conditions. They used PIV to observe the flow field immediately at the fan exits. The lowest flow speed is observed directly at the fan exit, increasing to a peak at 3 cm before decreasing with distance. A maximum local velocity of 7 m/s was observed.

As it can be seen, fans cannot effectively simulate jet impingement, and achieving a thinner profile is limited. Unlike fans that rotate a shaft, piezoelectric cooling devices can transfer momentum to fluids using oscillatory actuation in thin and scalable designs. This can be seen in new commercialized piezoelectric-based air pump systems that offer jet velocities of 52 m/s and are less than 3 mm thick [5]. Piezoelectric-based SJDs are another method of producing jet flows that demonstrate the capability of producing synthetic jets (SJs) of up to 250 m/s [6]. In recent years, SJs have found applications in active flow control, particularly in aerospace. These mechanisms are used in manipulating flow, altering boundary layer structure to induce favorable changes in aerodynamic forces on airfoils or vectoring jets [7–9]. Other applications are mixing enhancement [10,11], heat transfer from compact surfaces [12,13], spray vectoring [14,15], micro propulsion [16,17], and active control for miniature aerial vehicles [18].

Piezoelectric-based SJDs are structurally simple, lightweight, low-power, and dependable. When compared to natural convection, SJs improved local heat transfer by at least 3X, and in some cases exceeded 8X for small surfaces [19]. Further studies showed that SJ enhances the local heat transfer by up to 15 times compared to natural air convection [20]. Furthermore, the heat transfer characteristics of SJ with continuous jets (CJs) at the same boundary conditions showed better performance of stand-alone SJs. It was argued that the radial variation of time-averaged heat transfer for SJ was found to be 35 % better than for CJ [21]. A similar observation has been made by other researchers as well [22,23]. Arik et al. [24] reported 40 % higher heat dissipation for SJ as compared to the CJ for similar boundary conditions. Later studies claim that at the same range of Re number, SJs achieved further than twice the cooling effect of a CJ [25]. SJs also show better penetration into cross-flow. A comparison study between SJ's and CJ's cooling performance with the same net mass flow rate and identical jet outlet temperatures showed that without cross flow, CJ, and with cross-flow, SJ outperformed in terms of heat removal [26]. The most probable reason for the better thermal performance of SJs is the continuous formation of vortical structures and their sweeping effect upon impingement. Meanwhile, at low Re number CJs, vortex formation is not significant; hence, cooling is aligned with the streamlines forming on the surface.

Multiple comprehensive reviews have been published about the thermo-fluidic behavior of SJs. Efforts have been made to describe the effects of the orifice to plate distance, stroke length, frequency, orifice shape, jet vectoring, orifice plate thickness, cavity shape, and acoustics on the flow, and thermal characteristics of SJ based cooling [25,27–30]. Most of these articles call for additional research in this area. Manufacturing and prototyping of SJDs is likely to be one of the most significant challenges to conducting research in this field. This study presents an innovative approach to developing high velocity SJDs based on composite additive manufacturing, which is shown to be an appropriate rapid prototyping method. Manufactured SJDs are tested for deflection performance, and their acoustic and structural performance is documented and analyzed. To predict SJD output level, fluid dynamic-based lumped element models are tuned using deflection behavior only. Based on these models, the operation frequency of each SJD was calculated. Thermal tests were performed on devices at varying distances from the heated surface to assess their heat transfer capabilities. The jet output performance is then assessed using hot wire anemometry.

In the following sections of this article, a brief fundamental of SJDs, specifically piezoelectric SJDs, is discussed. The importance of impulse generation is highlighted in order to demonstrate that higher jet velocity cannot always be used to assess the performance of SJDs. Furthermore, coupled mechanical-Helmholtz resonator system of piezoelectric-based SJDs, phase lag between diaphragm and jet, and recent concerns about the accuracy of LEM models for thin profile SJDs are addressed. Methods for manufacturing piezoelectric-based SJDs and boundary conditions for actuators are discussed. A novel manufacturing method is developed that can accelerate SJD research by facilitating rapid prototyping of SJDs using additive manufacturing. The analytical, and experimental analyses are then presented, with additional details provided in supplementary documents.

2. Fundamentals of synthetic jets

An SJ is generated when pressurized fluid is rapidly expelled from a cavity through an orifice. This expulsion creates a vortex ring that propels the jet forward. As the jet moves away from the orifice, it entrains the surrounding air, maintaining its velocity and coherence due to the momentum carried by the vortex rings. Round orifices produce coherent vortex rings, resulting in focused and intense jet impacts, ideal for targeted heat transfer. In contrast, slot orifices create unstable vortex structures that disperse more rapidly, suitable for wider area cooling, or aerodynamic applications. Given the focus on concentrated heat transfer, this research will utilize round orifice SJs. To design an SJD for hotspot cooling three key issues need to be considered. First of all, an SJ needs to be formed (Strouhal/Stokes number criteria). Next, SJ should have high velocity, and at the same time have a high mass flow rate (balance between Re number, orifice diameter, and acoustic damping). Finally, SJD should be placed at an appropriate distance from the heated surface (Orifice-to-plate distance adjustment). It needs to be noted that an extension of this section can be found in the supplementary notes.

2.1. SJ formation

Fig. 1 provides an illustration of SJ formation. Considering the orifice as a cylinder with a diameter of D, and length of L_o, a cylindrical volume of fluid moves out of the orifice with varying radially averaged velocity $U(t)$ in each state of the ejection cycle (i.e., from $t = 0$ *to* $\tau/2$, while assuming jet is reaching its maximum velocity at $\tau/4$ where t is time and τ is the period for diaphragm deflection). It needs to be noted that there can be a phase difference between the oscillation of the diaphragm and velocity profile in the orifice, e.g., they can operate in opposite phases, as when there is a contraction in cavity volume, the fluid can be sucked in and when the cavity is expanding SJ can be ejected from the orifice.

The time (over ejection cycle) and area average velocity of ejected *puff* or *slug* ($\overline{U}$) (see Fig. 2(b)) can be evaluated as [31]:

$$\overline{U} = \frac{2}{\tau}\int_0^{\tau/2} U(t)dt = 2fr\int_0^{1/(2fr)} U(t)dt \tag{1}$$

Where fr is the frequency of diaphragm deflection. Equivalent length of the ejected slug (L_{es} – see Fig. 1(c)) can be defined as:

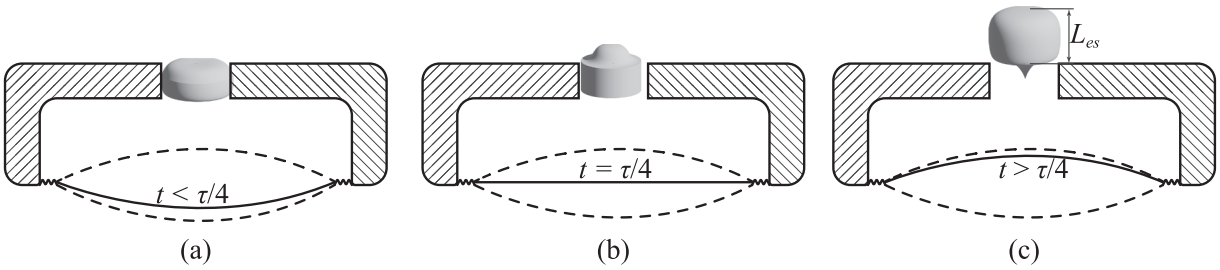


Fig. 1. The formation process of an SJ at (a) the beginning of the ejection cycle, (b) at the maximum velocity of the diaphragm, and (c) the end of the ejection cycle.

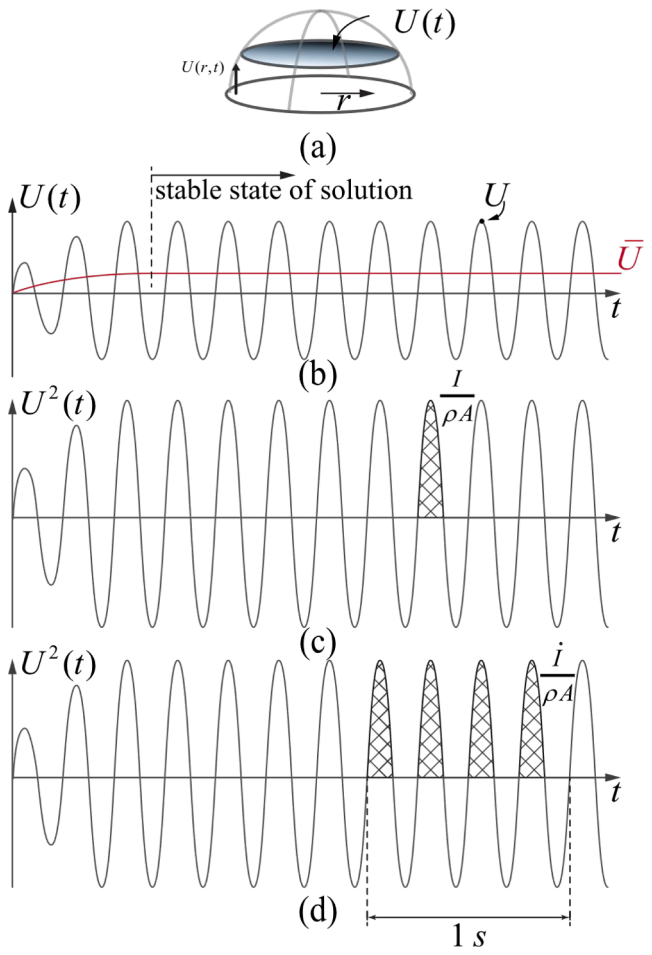


Fig. 2. (a) Visual description $U(t)$ as the radially averaged jet exit velocity $U(r, t)$, (b) behavior of $U(t)$ and visualization of its over ejection cycle average $\bar{U}$, and its maximum value U, (c) visualization of impulse generation per unit mass per unit area per cycle $I/\rho A$, and (d) its rate $I/\rho A$.

$$L_{es} = \bar{U}\tau/2 = \bar{U}/(2fr) \tag{2}$$

Now two dimensionless parameters for the SJ in the formation stage can be obtained as stroke length (L^*) and Reynold number [32]. Non dimensional stroke length is:

$$L^* = L_{es}/D \tag{3}$$

One can build a relationship to Strouhal number as:

$$St = \frac{fD}{\bar{U}} = \frac{1}{2(\bar{U}/(2fr))/D} = \frac{1}{2L^*} \tag{4}$$

Holman et al. [33] introduces Stokes number (Stk) as:

$$\mathrm{Stk} = \sqrt{2\pi fD^2/\nu_\infty} \tag{5}$$

Where ν_∞ is the kinematic viscosity in the ambient. Re number can be defined as:

$$\mathrm{Re} = \bar{U}D/\nu_\infty \tag{6}$$

One can see that:

$$\frac{\mathrm{Re}}{\mathrm{Stk}^2} = \frac{\bar{U}D/\nu_\infty}{2\pi fD^2/\nu_\infty} = \frac{\bar{U}}{2\pi fD} = \frac{L^*}{\pi} \tag{7}$$

Utturkar et al. [34] and Holman et al. [33] showed that by, empirical relationship SJ formation criterion-JFC, is:

$$\mathrm{JFC} = \frac{1}{St} = \frac{\mathrm{Re}}{\mathrm{Stk}^2} = \frac{L^*}{\pi} > K^* \tag{8}$$

Where K^* is jet formation constant depending on the geometry of the orifice shape, radius of curvature, and aspect ratio of the slot. K^* is 1 for slot orifice and 0.16 for circular orifice.

2.2. Impulse generation

From heat transfer point of view, it is well established that the Nu number is a function of Re number, which is also true in SJ impingement. In SJ impingement cooling, Nu number is correlated to Re^n with exponent n ranging from 0.492 [35] to 0.841 [36]. In CJ generation, one can achieve high Re numbers by reducing the nozzle diameter while compensating for the required pressure in the reservoir tank. This is not the case in low-inertia-diaphragm actuated SJDs, where the introduced damping in the orifice section can greatly influence the behavior of the low-inertia-diaphragm (usually a voice coil or a piezoelectric diaphragm). Since the design of such devices typically centers around the actuation mechanism, Re number cannot simply be increased by tightening the orifice.

Thus, reducing the orifice diameter cannot always lead to higher Re numbers, as it can significantly damp the actuator. Conversely, significantly widening the orifice can reduce the Reynolds number, potentially failing the device to meet the JFC criteria in Eq. (8). As a result, the design should achieve a balance. When designing SJDs for heat transfer applications, as shown later in the results, it is necessary to consider both the Reynolds number and impulse generation (I), as well as their effects on diaphragm behavior. The hydrodynamic impulse per unit width per cycle can be defined as (see Fig. 2(c)):

$$I = \int_0^{\tau/2} (\rho_\infty A U(t)) U(t) dt = \rho_\infty A \int_0^{\tau/2} U^2(t) dt \tag{9}$$

In some article [27,37] one can see that right hand side of Eq. (9) is being integrated as $I = \rho A \bar{U}^2 \tau/2$ which can be considered oversimplification since $U(t)$ is a function of time. In this article, we introduce a new term to assess the behavior of SJDs as hydrodynamic impulse *rate* per unit width, I. I can be defined as (see Fig. 2(d)):

$$\dot{I} = I \cdot fr \tag{10}$$

This is an important parameter that contains valuable data for SJD performance evaluation. For example, if an SJD generates high velocities but has a small orifice area, I will be low. If the orifice area is large, the velocity can decrease, resulting in a reduction of I and I. Additionally, the frequency determines the rate of momentum ejection into the system. Thus, an SJD is superior if it can deliver a high velocity jet through a larger orifice at a higher frequency. Because most low-inertia-diaphragm-actuated SJDs meet JFC criteria at their resonance modes, they should be manufactured in such a way that a balance between A, and U at these resonance frequencies exists.

Fig. 2 provides a visual aid in understanding $U(r,t)$, $U(t)$, U, $\bar{U}$, I, and I. $U(t)$ is the radially averaged jet exit velocity of $U(r,t)$. U is the maximum achievable $U(t)$, and $\bar{U}$ is over ejection cycles averaged $U(t)$. By considering constant density, I can be conceptualized as proportional to the area under $U^2(t)$ curve in every ejection cycle, and I can be conceptualized as proportional to the area under $U^2(t)$ curve in every ejection cycle that can happen in one second.

2.3. Impingement characteristics

SJ impingement is similar to jet impingement, with two major distinctions. First, a continuous impingement of generated vortical structures occurs on the heated plate. Second, the temperature of the SJ can be affected and fluctuate depending on the distance between the orifice and the heated plate due to strong mixing, and the direction of gravity driven convection on the heated plate [26]. After impingement, vortical structures advect in the radial direction, and induce counter-rotating secondary vortices in the wall jet region [38,39]. However, it is shown that in high spacing, and also very low spacing to the heated plate, secondary vorticities can be diminished [40].

It was also demonstrated that small stroke length caused vortex rings to take longer to reach the wall, resulting in a decrease in vortex strength. Weak impingement resulted in the formation of large-scale spiral vortices in the wall-jet region, whereas vortex rings with long stroke lengths were found to be strong enough to generate strong secondary vortex rings in the wall-jet region. The findings predict that smaller stroke lengths are advantageous for heat transfer due to the formation of large-scale spiral vortex rings close to the wall. [41]. In contrast, in another study, it was observed that the heat transfer performance increased linearly with the increase in stroke length [42]. There are multiple other disparities when trying to quantify Nu number of SJ impingement, or provide correlations [28]. The main reason can be attributed to the fact that SJ impingement is more complex in comparison to CJ impingement. Numerous factors influence Nu, such as Re, Stk, placement, shape, and strength of the vortical structure, characteristic of the heated surface, shape of orifice, geometrical and material selection of SJDs, confinement of impingement by SJD, boundary conditions, natural convection, and the self-heating effect of SJD. All of these parameters raise the uncertainty of Nu number prediction and generalization of SJ impingement heat transfer.

Probably, leveraging the Sherwood number would be more beneficial. Techniques such as the Naphthalene sublimation study [43] allow for more manageable experimental set ups.

In summary, the distance to the heated plate is what all studies agree on. Most studies have found an increasing and decreasing trend in the orifice-to-surface distance [23,44,45]. Thus, it can be stated that for any SJD impingement set up, there is an optimal impingement distance where heat transfer rates are maximum.

2.4. SJD behavior

Fig. 3 depicts a typical simplified body diagram for a piezoelectric-based SJD and its conceptualization using the Lumped Element Model (LEM). Details of the LEM model is provided in supplementary notes.

LEMs can provide a swift understanding of SJD's dynamic response [46]. Sharma [47] introduces a fluid dynamics based model which is

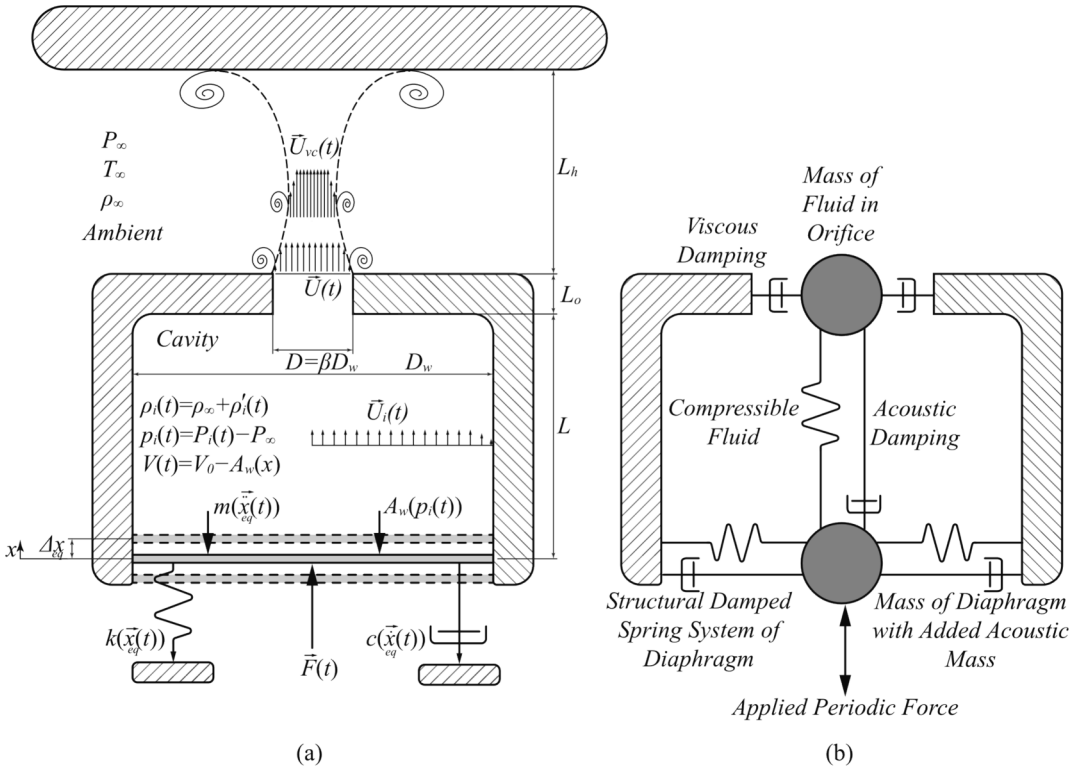


Fig. 3. (a) Schematics of the studied SJD with body diagram, and (b) conceptualization of the SJ actuator as a coupled mechanical-Helmholtz resonator system with two degrees of freedom.

summarized herein. Driving force $F(t) = f(v(t), structure)$ can be conceptualized as a function of input voltage $v(t)$ and structure of the piezoelectric actuating mechanism. As shown in Fig. 3(a), after containing this actuator inside a cavity with an orifice, apart from the mass and stiffness of the actuator itself, acoustic mass and stiffness also is added to it [48,49]. Thus the actuator will have a total mass $m = m_w + m_f$ which includes the mass of the oscillating wall m_w(mass of shim, piezoelectric ceramic, and solders) and added an acoustic mass of fluid inside the cavity m_f [50]. The damping coefficient (c) also consists of two components: the structural damping of the oscillating wall (c_w) and the acoustic radiation damping factor (c_f). The latter results from the energy loss carried away by sound waves generated during vibration. In the context of a circular piston vibrating within an infinite baffle at a radian frequency of $\omega = 2\pi fr$ the added acoustic mass and damping terms are [47]:

$$\begin{aligned} m_f &= \frac{2\rho_\infty A_w U_{sound} H_1(y)}{y\omega} = \frac{4\rho_\infty A_w U_{sound}}{\pi\omega} y\left(\frac{1}{3} - \frac{y^2}{3^2\cdot 5} + \frac{y^4}{3^2\cdot 5^2\cdot 7} - \cdots\right) \\ c_f &= \rho_\infty A_w U_{sound}\left(1 - \frac{2J_1(y)}{y}\right) = \rho_\infty A_w U_{sound} y^2\left(\frac{1}{2\cdot 4} - \frac{y^2}{2\cdot 4^2\cdot 6} + \frac{y^4}{2\cdot 4^2\cdot 6^2\cdot 8} - \cdots\right) \\ y &= \frac{\omega D_w}{U_{sound}} \end{aligned} \tag{11}$$

where ρ_∞, A_w, U_{sound}, H_1, J_1, and D_w are unperturbed fluid density (initial density, or density outside the cavity), the surface area of the oscillating wall, speed of sound, Struve function of the first kind, Bessel function of the first kind, and the diameter of the oscillating wall (herein we assume its equal to the diameter of the cavity), respectively. Moreover, speed of sound is:

$$U_{sound} = \sqrt{\gamma \frac{P_\infty}{\rho_\infty}}\ ,\ \gamma = C_p/C_v \tag{12}$$

where γ, C_p, C_v, and P_∞ are ratio of specific heats ($\approx$1.4 for air), specific heat in constant pressure, and constant volume, and initial cavity pressure (or pressure outside the cavity), respectively. One can see that m_f is in direct relation to D_w^3, while acoustic damping is in direct relation to D_w^4 and also ω^2.

During actuation, the gauge pressure inside the cavity, $p_i(t)$, undergoes fluctuations around a zero-mean value as the fluid experiences compression and expansion. It is crucial to highlight that these pressure fluctuations are negligible compared to the overall pressure inside the cavity, allowing LEMs to assume that the cavity pressure at any given moment is approximately equal to the pressure outside the cavity ($P_i(t) \approx P_\infty$). Referring to Fig. 3(a), an equation governing the dynamics of the flexible membrane can be derived as:

$$m\ddot{x}_{eq}(t) + c\dot{x}_{eq}(t) + kx_{eq}(t) = F(t) - p_i(t)A_w \tag{13}$$

in which $A_w = \pi D_w^2/4$. It should be noted that in LEM, diaphragm deflection is simplified to piston cylinder motion as defined by x_{eq}. Later, it will be discussed how the piston cylinder motion relates to the actual deformation of the piezoelectric disc diaphragm x. x is the displacement in the center of the deflecting disc.

Equation (13) governs the characteristics of the actuation of the wall. It needs to be noted that when the force is applied in a period with an angular frequency of ω as $F(t) = F_{max}\sin(\omega t)$, the deflection is no longer constant and varies with the frequency of the applied force. Considering a simple damped spring mass system with a damping ratio of ζ and an undamped natural frequency of ω_n one can write [51]:

$$\Delta x_\omega = \frac{\Delta x_{static}}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\zeta\frac{\omega}{\omega_n}\right)^2}} \tag{14}$$

To estimate the deflection in the angular frequency of ω, Δx_ω, multiply static deflection by the dynamic amplification factor. This is a simplified analysis that shows the difference in deflection behavior between static, and dynamic loads. Furthermore, it should be noted that a damped spring mass system will typically provide three distinctive frequencies: undamped natural frequency, damped natural frequency, and frequency of maximum force amplitude. The damping ratio for the wall can be defined as $\zeta = c/2\sqrt{mk}$. Because the damping ratios for thin metal plates are low (≤ 0.03), these three frequencies are very close and can be considered the same. Thus, one can use single wall resonance of $\omega_w = \sqrt{k/m}$. ζ and ω_w are structural characteristics of the diaphragm.

By assuming $\zeta \cong 0.02$–0.03 one can determine the resonance frequency of the diaphragm by measuring its displacement in different frequencies. Knowing value of $m \cong m_w$ one can experimentally determine k as $k = \omega_w^2 m$. Next, F_{max} can be evaluated. By driving the actuator at a low frequency such that $\Delta x_\omega \cong \Delta x_{static}$ (Refer to Eq. (14)) one can estimate $F_{max} = k(\Delta x_\omega)$. In summary, the diaphragm's stiffness is determined by its resonance frequency, and the applied force is obtained by driving the device at a low frequency (here at 25 Hz).

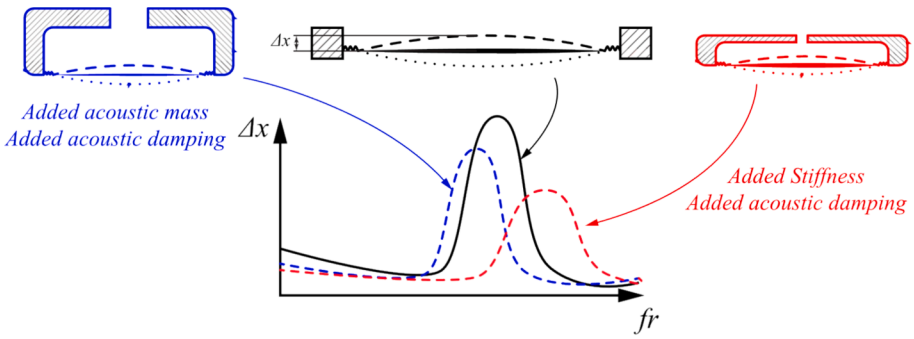
When the diaphragm undergoes oscillation, the fluid inside the cavity experiences alternating compression, and expansion. Simultaneously, there is an oscillatory flow through the orifice. Sharma [47] shows the second resonance of the system contributes to flow behavior through the following equation:

$$\ddot{U} + \frac{1-\beta^4}{L_e C_d^4}|U|\dot{U} + \omega_h^2 U = \frac{A_w}{A}\omega_h^2 \dot{x}_{eq} \tag{15}$$

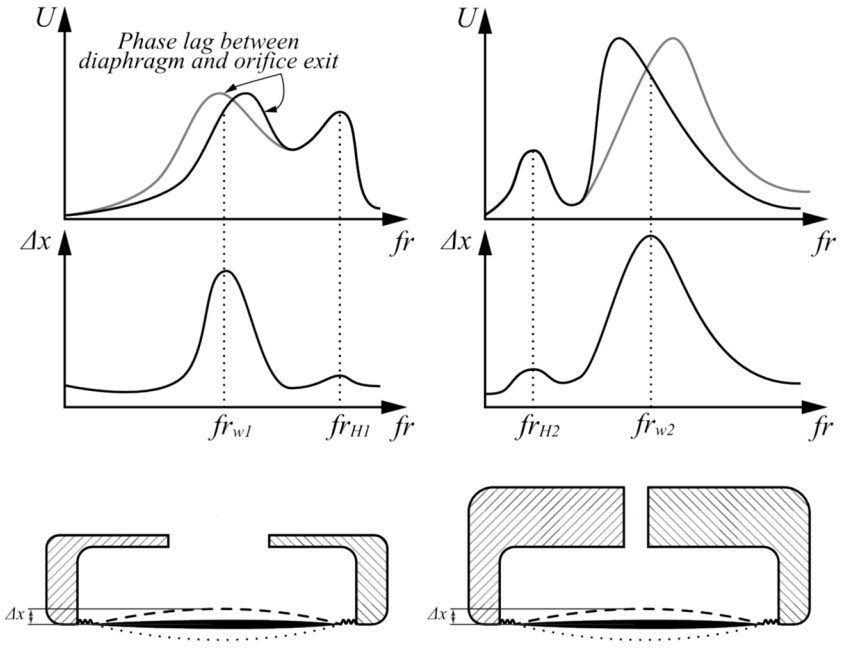
where $\beta = \frac{D}{D_w}$, $C_d = f(\beta, Re) \cong 0.6$ is viscous loss coefficient, and $L_e = L_o + C_l\sqrt{A}$ is the effective length of the air jet/slug at the orifice, and $\omega_h = \sqrt{k_h/m_h} = \sqrt{(\gamma A^2 P_\infty/V_0)/(\rho_\infty A L_e)} = U_{sound}\sqrt{A/(V_0 L_e)}$ represents the Helmholtz frequency. L_o is the length of the orifice and C_l is the inertia coefficient which needs to be tuned for each LEM model.

Equations (13), and (15) show that SJD can be conceptualized as a coupled mechanical-Helmholtz resonator system with two degrees of freedom. This implies that the actuation mechanism has two resonance modes, and maximized output can be achieved in *near* resonance frequencies.

Fig. 4 depicts the behavior of SJD provided in equations discussed in this section. These illustrations are based on studying the LEM of Sharma [47] in different geometrical inputs. Fig. 4(a) provides the deflection behavior (Δx, or Δx_{eq}) with respect to operating frequency (fr) of the diaphragm at a constant driving voltage. Deflection gradually reduces by increasing frequency until a significant deflection can be seen in the first principal wall resonance frequency (fr_w). Adding such a diaphragm to a cavity in the form of a SJD results in two major changes. (1) First, added acoustic mass to the diaphragm can reduce fr_w to a lower level in



(a) *Effect of Cavity on Diaphragm Behavior*



(b) *Illustration of Coupled Mechanical-Helmholtz Resonator System*

Fig. 4. (a) Diaphragm deflection behavior before and after adding cavity. (b) Diaphragm deflection and SJ output behavior of SJD showcasing higher Helmholtz resonance on left, and lower Helmholtz resonance on the right.

comparison to vibration without cavity (blue curve), while added stiffness can increase fr_w (red curve). (2) Peak deflection in fr_w also reduces due to acoustic damping, which damps the vibration of the diaphragm. After adding the cavity, a slight increase in deflection at Helmholtz resonance (fr_H) can be seen. The closer the Helmholtz and first principal resonance frequencies are together, or the lower the L value (thinner the cavity), the greater the deflection in Helmholtz resonance. Ultimately, by optimizing the SJD, either by changing the size of the cavity or orifice, or by modifying the mass of the diaphragm or its stiffness, the Helmholtz and wall resonance frequencies can be brought closer to the same value, resulting in a significant increase in the system output. This can be conceptualized as mode coupling, where SJD will produce its highest jet output velocity.

Fig. 4(b) provides the deflection behavior (Δx, or Δx_{eq}) and instantaneous area averaged jet exit velocity (U) vs. driven frequency (fr) of the diaphragm in constant voltage level applied to the piezoelectric bender. It can be seen that by modifying the characteristics of the orifice, fr_H can occur in higher or lower frequencies than that of diaphragm resonance. Depending on this, phase lag existing between diaphragm and orifice the maximum jet exit velocity will occur at a slightly lower or higher frequency than the wall resonance frequency.

In some cases, as the cavity of SJD becomes thinner and deviates from an ideal Helmholtz resonator, Helmholtz resonance theory stated to be deviating from experimental measurements by more than 130 % [52]. It is suggested that in thin cavities, a quarter-wave resonance model be used which aligned much closer to the observed data, with discrepancies around 15 % [52]. However, it should be noted that these conclusions were drawn using a different type of SJD that lacks axial symmetry and is not the same as the type of SJD used to derive LEMs. In other words, this type of conclusion is derived from SJD type (*b*), as illustrated in Fig. 5. Our experiments on type (*a*) SJD showed good accuracy of the LEM model. However, more experiments would be useful for a generalizing solution.

A recent study [53] suggests that the characteristic resonance frequency of the SJD should be evaluated first based on the pressure acoustic FEM model and replace ω_h in Eq. (15). The combination of replacing the Helmholtz frequency with a more accurate characteristic frequency and using computed loss coefficients resulted in more accurate LEM performance predictions for low-profile SJDs (Type b in Fig. 5 (b)). However, this consideration undermines the first major assumption used in LEM models, so it begs the question of whether the model is still accurate if the acoustic resonance mode is changed. This assumption, which serves as the foundation for LEM models, states that the internal pressure in the cavity is uniform at any given time (except near the

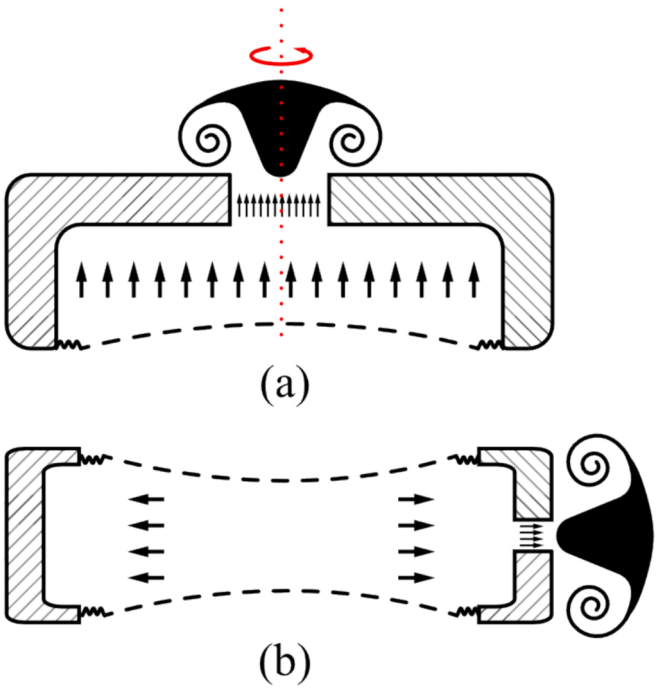


Fig. 5. (a) Type a SJD where a sort of axial symmetry exists and LEM models are based on, and (b) type b SJD in which an orifice is introduced from the side.

orifice). This is because it was considered that the wavelength of the pressure oscillations is significantly larger than a typical dimension L of the actuator cavity. This condition ensures that the frequency of the most significant standing wave of quarter-wavelength is well beyond the operating frequency range of the actuator. This can be shown by analyzing a wave equation describing a standing wave field in one dimension (x) with a simple order of magnitude simplification [50]:

$$\frac{\partial^2 p}{\partial x^2} = \frac{1}{U_{sound}^2}\left(\frac{\partial^2 p}{\partial t^2}\right) \xrightarrow[\text{simplification}]{\text{order of magnitude}} \frac{\Delta p}{L^2} = \frac{1}{U_{sound}^2}\left(\frac{p}{1/f^2}\right) \rightarrow \Delta p = \frac{L^2}{\lambda^2} p \tag{16}$$

This consideration seems to be true in type (a) SJD as shown in Fig. 5(a) since the orifice is considered to be in front of the diaphragm. However, when the orifice is located on the side of the diaphragm, "L" in Eq. (16) needs to be replaced by "D_w". In this case, it is obvious that the pressure inside the cavity will not be uniform (see Fig. 5(b)) and LEM models can be inaccurate.

3. Development of ARTgroup SJD Concept, and manufacturing

3.1. Actuation mechanism

One important aspect of manufacturing an SJD is to generate an adequate oscillatory actuation. Piezoelectric actuation is a promising way to create SJDs. The piezoelectric composite diaphragms used in SJDs are typically circular metallic shim discs, often made of brass/steel, with a piezoceramic patch attached to one or both sides. Chen et al. [54] provide a detailed roadmap for manufacturing such benders. However, due to the high cost of manufacturing, the piezoelectric diaphragms in most research cases are sourced from commercially available units, also known as piezoelectric disc benders, which have industrial applications in horns, tweeters, and sirens. These piezoelectric ceramics are typically made of conventional zirconate-titanate PZT-5A/5H. The electromechanical coupling ratio of conventional polycrystalline piezoceramics is low. Recent research [55] has shown that more electromechanically efficient piezoceramics, such as single-crystal lead magnesium niobate-lead titanate (PMN-PT), can achieve superior performance metrics. The PMN-PT-based SJD achieved a peak exit jet velocity of 99.5 m/s at a peak supply voltage of 40 V, with a power conversion efficiency of 72 % at the SJD's Helmholtz frequency.

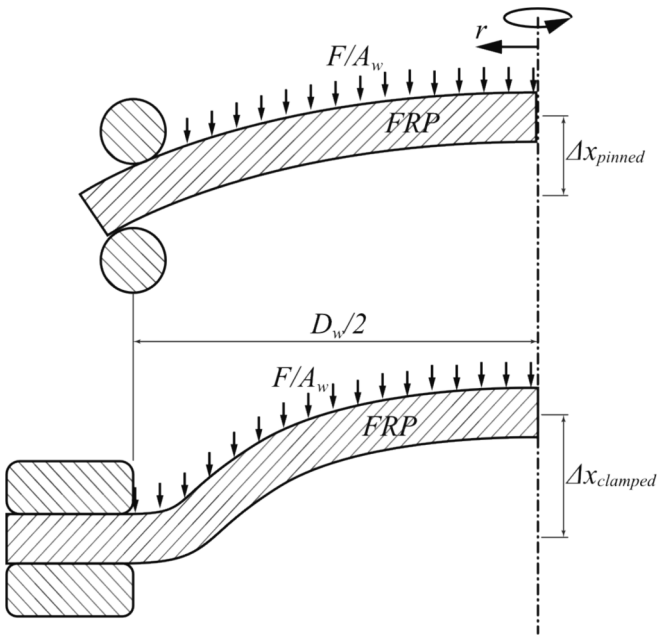


Fig. 6. Pinned (up) and clamped (down) boundary conditions for discs with known flexural rigidity of FRP with diameter D_w and surface area A_w, with central deformation Δx under constant force (F).

3.2. Boundary condition and displacement of piezoelectric diaphragm

Manufacturing a cavity and orifice is straightforward. It can be done through the machining process or additive manufacturing. The main question is how to add piezoelectric based actuator to it. A structure is required to hold the diaphragm in a sealed manner to the cavity, while giving it an ability to deform. Depending on it, the diaphragm will behave as if being clamped [56], pinned [54], as shown in Fig. 6, or in between the two.

Pinning allows for more deflection, whereas clamping reduces deflection while providing a stiffer structure with a higher resonance frequency. Timoshenko [57] demonstrates that for circular plates with known flexural rigidity of *FRP*, Poisson's ratio of N, and uniform applied force of F/A_w, the static deflection ratio from pinned to clamped is equal to $(5 + N)/(1 + N)$. For $N = 0.3$, the defection in the center from the pinned case is approximately four times the clamped case. Thus, the pinned case can be considered the best method for piezoelectric-based actuation in SJDs. Fig. 7 shows various types of piezoelectric-based SJD designs. Fig. 7(a) and 7(c) show examples of clamped and pinned boundary conditions applied to a cavity, as previously discussed.

An alternative piezoelectric assembly method for producing SJDs, suggested by Garg et al. [61], involves using two piezoelectric bimorph disks separated by a flexible wall material with an orifice, named advanced piezoelectric dual cooling jet (DCJ). In such configurations, piezoelectric benders are attached at the edges with flexible adhesive (typically silicone). The boundary condition of the piezoelectric disc falls in between clamped and pinned conditions [50]. The silicone adhesive could absorb and dissipate vibrational energy, reducing vibration amplitude. This was recently demonstrated in a comparison of clamped boundary conditions to flexible boundary conditions using neoprene rubber washers, where neoprene rubber washers reduced deflection amplitude [62]. Research on such SJDs can also be found in [23,63–65]. The commercialization of these devices began with research by Garg et al. [20], which studied DCJs operating at resonance frequencies

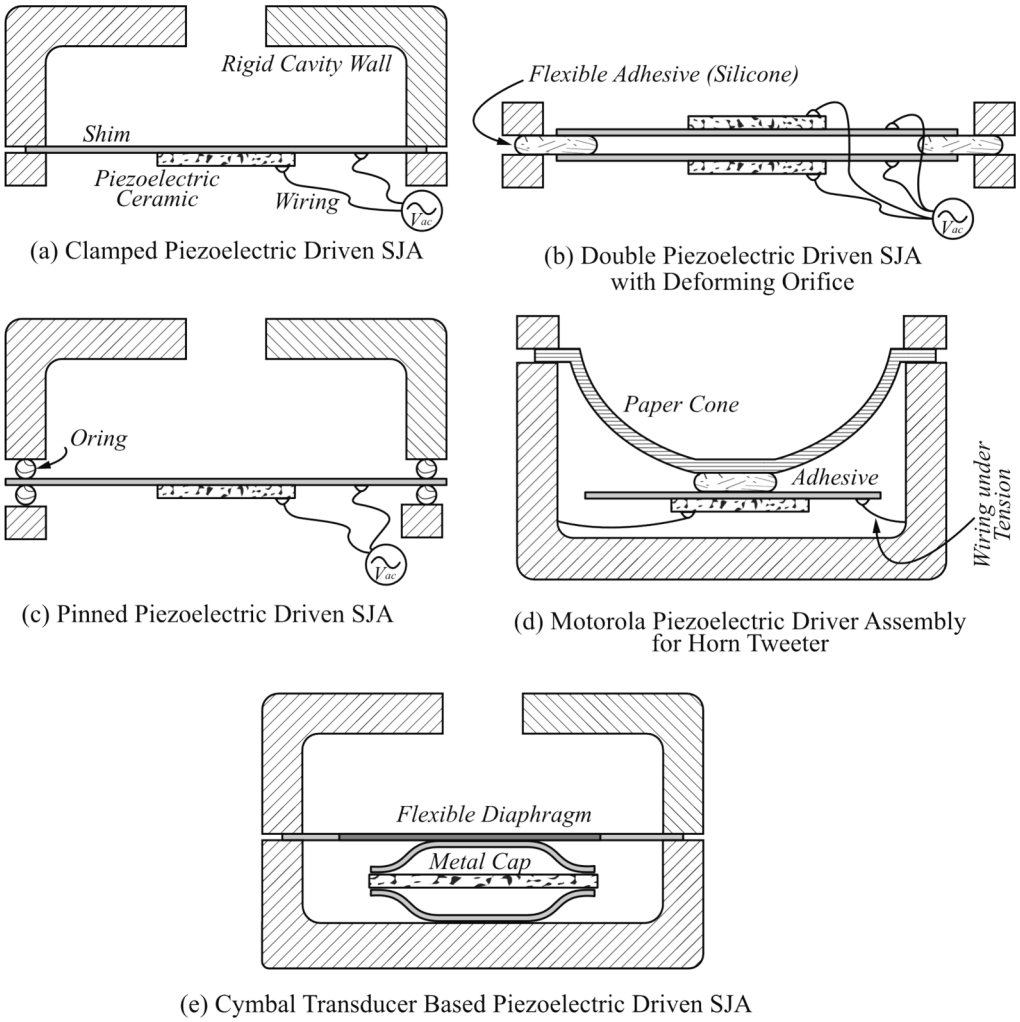


Fig. 7. Various deforming diaphragm mounting configurations for small scale SJDs. (a) Clamped piezoelectric bender on edges, e.g., Chiatto et al. [56], (b) clamped together piezoelectric benders by flexible adhesives and also suspended by the same flexible adhesive, e.g., De Bock et al. [58] (c) pinned piezoelectric bender, e.g., Chen et al. [54], (d) suspended piezoelectric bender, e.g., Bost [59], and (e) Piezoelectric cymbal transducer based SJD, e.g., Ayaskanta et al. [60].

between 3000 and 4500 Hz. These were able to achieve jet exit velocities above 90 m/s with 110 V peak to peak applied to circular piezoelectric benders, and a heat transfer coefficient of approximately 239 W/m^2-K was reported. Subsequently, to reduce noise levels for consumer cooling applications, thinner and larger square-shaped piezoelectric discs were used to lower the resonance frequency to 150–175 Hz, with a total active device thickness of 1 mm [58]. As a result, velocities dropped to less than 4 m/s, with an effective heat transfer coefficient as high as 33 W/m^2-K.

Using two piezoelectric diaphragms, similar to the configuration shown in Fig. 7(b), demonstrated improved performance if the piezo disc was clamped by rigid structures and had a solid orifice shape. Such designs are known as pancake SJDs, and Van Buren et al. [66] have extensively studied them. The dual disc configuration resulted in a 40 % increase in jet velocity compared to the single disc configuration. The SJ reached a peak velocity of 211 m/s from a 12 × 1 mm^2 rectangular orifice at 700 Hz, indicating significant momentum transfer compared to other designs in the literature.

An alternative approach to utilizing piezoelectric disc benders is depicted in Fig. 7(d), showcasing a suspended type of piezoelectric diaphragm configuration. This method was first introduced by Motorola in 1975 for their tweeter horns [59]. This design bears a resemblance to speakers, wherein the piezoelectric disc bender is suspended by being centrally glued to a paper cone. Additionally, two flexible wires are soldered to the disc, aiding in keeping the piezoelectric disc suspended alongside the paper cone. It should be noted that this particular design can be used for SJD and is tested in our laboratory.

The cymbal transducers are novel actuators that reduce stress concentration on piezoceramic disks while increasing displacement. This transducer configuration has a piezoelectric ceramic disk sandwiched between two metal end caps. A vertical load applied to these end caps causes radial stress in the piezoceramic. Because of the high cost, the use of this configuration is scarce. Ayaskanta et al. [60] used a method similar to Fig. 7(a) to produce SJ with peak velocities around 35 m/s from a 0.9 × 0.9 mm^2 orifices at a resonance frequency between 700–750 Hz.

3.3. SJD manufacturing with micro carbon fiber filled polymer 3D printing

To conduct this study, SJDs based on the concepts shown in Fig. 7(a), (b), (c), and (d) are manufactured and tested. The design resulted in bulky designs, complex manufacturing, or a lack of control and manipulation over cavity and orifice characteristics. As a result, a new method is proposed based on the power of 3D printing. Next subsections explain the novel manufacturing process.

Table 1
Piezoelectric Diaphragm Properties.

Plate material	Plate diameter	Capacitance	Brass thickness	Ceramic Thickness	Ceramic Dia	Mass	Nominal Resonance	Manufacturer
Brass	44 (mm)	70 (nF)	0.1 (mm)	0.13 (mm)	25 (mm)	5.1 (g)	600 (Hz)	CUI Devices/CEB-44D06 [67]

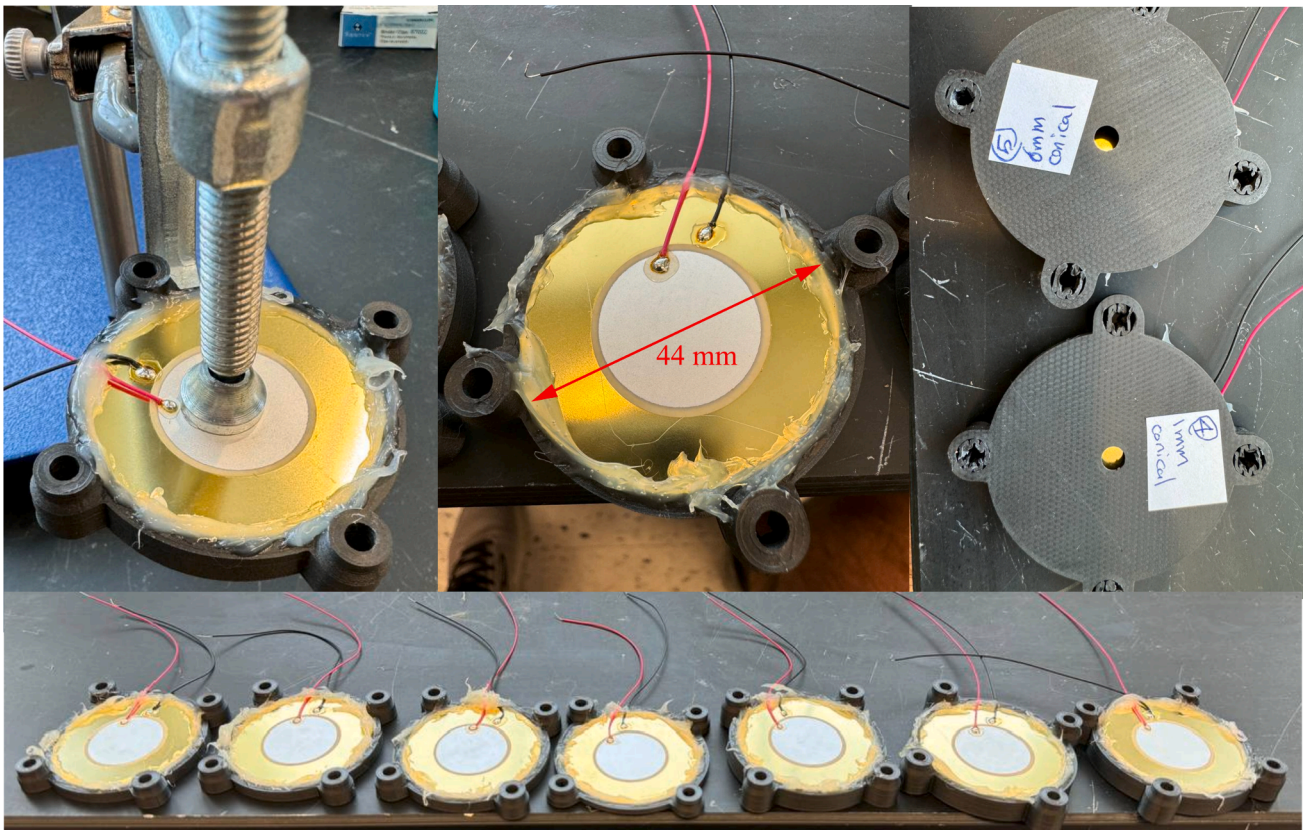


Fig. 8. Placing piezoelectric bender disc designed edge of the SJD body with a temporary clamp and applying adhesive.

3.3.1. *Initial devices*

Table 1 shows the diaphragm's characteristics. The diaphragm is initially powered at a peak-to-peak voltage (V_{pp}) of 140 V_{pp}. However, it was realized that in continuous operation with Δx at the center of the -disc exceeding 400 μm, ceramic fails. Thus, experiments are performed at 100 V_{pp}.

To achieve a boundary condition similar to the pinned installation of the piezoelectric disc (see Figs. 6, and 7(c)), a new method was developed following extensive prototyping and trial and error to determine appropriate materials. This method was developed to ensure that the piezoelectric disc can be held with the least amount of weight and smallest structural footprint while maintaining maximum deflection and reliability in resonance actuation. The manufacturing process is illustrated in Fig. 8. The Markforged X7 industrial carbon fiber 3D printer with 50 μm resolution is used to print the body of the SJD, including the orifice, cavity shape, and supporting structures. The piezoelectric bender disc is then placed on a 45° edge designed to match its diameter and secured with a temporary clamp. The disc is then attached to the 3D printed structure with acrylic adhesive. Used acrylic adhesive has a shear strength of 20.7 to 26.2 MPa, a tensile strength of 20.7 to 27.6 MPa, and a tensile elongation of 5–15 %. These properties provided a strong bond between the shim and the cavity, temperature resistance, and flexibility for fatigue resistance. Not only was reliable operation achieved in the extended-duration studies, but also exceptionally high deflection amplitudes were obtained.

As it can be seen in Fig. 9, two types of cavities are considered (a) conical, and (b) cylindrical. Total of seven SJD bodies are manufactured for the first set of studies (Fig. 9(c)). SJD-1 has conical cavity $D_o = 4.75$ mm, $L_o = 0.5$ mm, and $L = 3$ mm (see Fig. 3(a)). SJD-2 is similar to SJD-1, but has a cylindrical cavity. SJD-3 is similar to SJD-1 but has a wider orifice $D_o = 6.75$ mm. SJD-4 is similar to SJD-1, but has a thinner cavity $L = 3$ mm. SJD-5 is similar to SJD-1 but has a larger cavity $L = 6$ mm. SJD-6 is similar to SJD-1, but has a larger cavity $L = 6$ mm, and a longer orifice $L_o = 1$ mm. SJD-7 is similar to SJD-5 but has a cylindrical cavity. Cavity, and orifice parameters are shown in Fig. 3 (a). However, to define the body of the SJD, additional parameters are introduced in Fig. 9 (a) and (b). L_t is the perpendicular length of the 45-degree edge on which the piezoelectric disc rests. L_T is the total length of the SJD. D_t is the required thickness for the SJD's sides so it can provide enough space for applying adhesive. The SJD body's total radius is defined as $D_T + D/2$. Details of all seven SJDs can be seen in Table 2. It can be seen that the cavity volume ratio to SJD-1 is also provided. This shows that using conical cavity instead of the cylindrical cavity can help reduce the cavity volume by almost two times. This allows one to change the Helmholtz resonance, reduce the acoustic mass in the cavity, and direct the flow toward the orifice.

3.3.2. *Optimized geometry*

As shown in Fig. 4(b), there is a gap between Helmholtz resonance and wall structural resonance when SJD is not optimized to match these

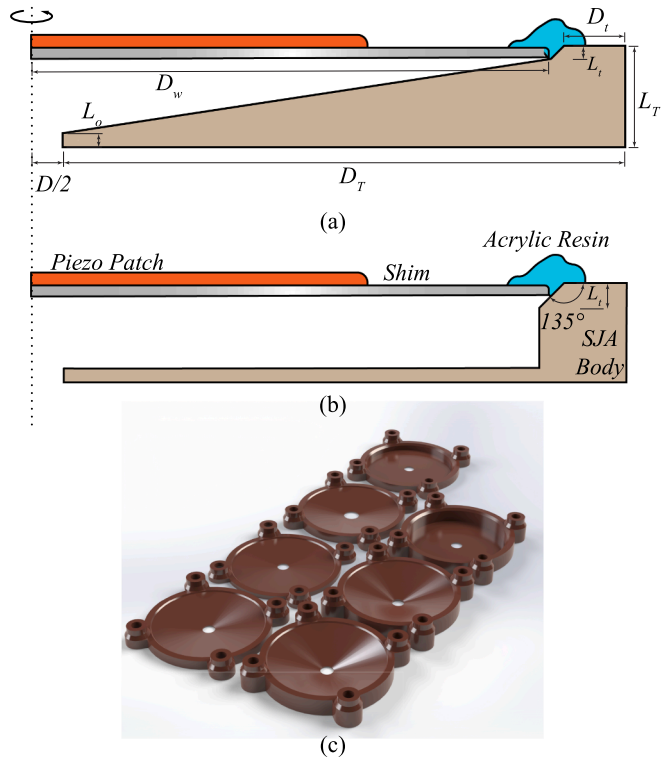


Fig. 9. Axisymmetric cut plane view of manufactured SJDs with (a) conical cavity, and (b) cylindrical cavity showcasing placement of piezoelectric disc on 45° edge and maintaining the structure using acrylic resin to provide pinned boundary condition.

two resonances. By changing the structural design of the cavity and orifice so that these two resonances match, one can see a significant change in jet exit output. By using parametric study with tuned LEM, $D = 1.65$ mm and $L_o = 2.3$ mm are set. Two SJDs are designed and manufactured and are named SJD 8-con and SJD 8-cyl. Details of optimized SJDs can be seen in Table 3.

4. Experimental study

The experimental procedure of this research is illustrated in Fig. 10. Initially, a suitable piezoelectric-based diaphragm was selected. Displacement measurements were performed using laser displacement sensors at a voltage of 140 V_{pp}. However, during prolonged operation at this voltage, the high displacement of the diaphragm caused the piezoceramics to crack, with SJDs 1, 2, and 7 failing. Results of the 140 V test are presented in supplementary notes. The primary reason for failure was not voltage breakdown but excessive displacement, which the ceramics could not withstand. Other aspects of the design, such as adhesive joints and the body of the SJDs, endured long-term operation. Subsequently, the voltage was reduced to 100 V_{pp}, at which level the SJDs operated safely.

Experimental displacement data were then extracted and used to tune LEMs for each SJD. Force (F) and stiffness (k) were evaluated based on low frequency operation data (see Eq. (14), and C_l was tuned to match displacement curves achieved from displacement data to LEM data. By matching displacement data from with the LEM predictions, we could confidently predict the velocity behavior and identify the frequency at which the SJDs would produce the highest jet exit velocity. The LEMs were subsequently used for a parametric study to match two resonance modes ($fr_H = fr_w$) based on parametric optimization of orifice characteristics. Furthermore, the LEM provided tools to evaluate integration in Eq. (9) and Eq. (10).

A thermal experimental setup was designed and constructed, with details provided in the supplementary note. The SJDs were tested at their evaluated frequencies, where they provided the highest jet exit velocity across various orifice-to-impingement plate distance ratios. Lastly, hotwire anemometry tests were performed to verify the accuracy of the tuned LEMs in predicting SJD behavior based on their geometry and diaphragm characteristics. This validation can help future research confidently optimize devices without the need for expensive and fragile hotwire test setups, as displacement measurement techniques are significantly more cost-effective.

4.1. Deflection measurement and LEM tuning

The deflection test is performed using Acuity AR100 laser triangulation sensor that measures distance by projecting a beam of laser light that creates a spot on a target surface. Diffuse reflected light from the surface is viewed from an angle by a CMOS detector array inside the AR100 sensor. The target's distance is calculated from the image pixel data using the sensor's microprocessor. The distance is transmitted through serial communications. The range of the measurement with the current unit is 10 mm (focal point at 5 mm), with resolution, of 0.01 % giving the nominal accuracy of 1 μm. The laser operates at 660 nm. The data acquisition frequency is 9400 Hz. Real world accuracy of the measurements was determined to be 3 – 5 μm, due to other vibrations, the orientation of the surface with the laser beam, and centering the beam. As seen in Fig. 11, a test bed is prepared to secure each SJD underneath the laser displacement sensor. SJDs are secured using four bolts to the test bed. The center of the disc is marked with a black marker, this will also help reduce the specular reflectivity of the silver coating on the disc. SJDs are being controlled using a custom LabView program. The

Table 2
Details of seven manufactured SJDs. See Fig. 3(a) and Fig. 9 (a) and (b) for parameter definition. (con for conical, and cyl for cylindrical).

SJD #	Cavity	D (mm)	D_w (mm)	D_t (mm)	L (mm)	L_o (mm)	L_t (mm)	L_T (mm)	V_0 (mm^3)	Volume ratio to SJD 1
1	con	4.75	44	22.6	3	0.5	0.5	4	2307	1.00
2	cyl	5	44	22.6	3	0.5	1	4.5	4562	1.98
3	con	7	44	21.6	3	0.5	0.5	4	2339	1.01
4	con	5	44	22.6	1	0.5	0.5	2	770	0.33
5	con	5	44	22.6	6	0.5	0.5	7	4620	2.00
6	con	5	44	22.6	6	1	0.5	7.5	4620	2.00
7	cyl	5	44	22.6	6	0.5	1	7.5	9123	3.95

Table 3
Details of optimized manufactured SJDs. See Fig. 3(a) and Fig. 8 (a) and (b) for parameter definition. (con stands for conical, and cyl stands for cylindrical).

SJD #	Cavity	D (mm)	D_w (mm)	D_t (mm)	L (mm)	L_o (mm)	L_t (mm)	L_T (mm)	V_0 (mm^3)	Volume ratio to SJD 1
8	cyl	1.65	44	22.6	3	2.3	1	6.3	4562	1.00
8	con	1.65	44	22.6	3	2.3	0.5	5.8	2284	0.99

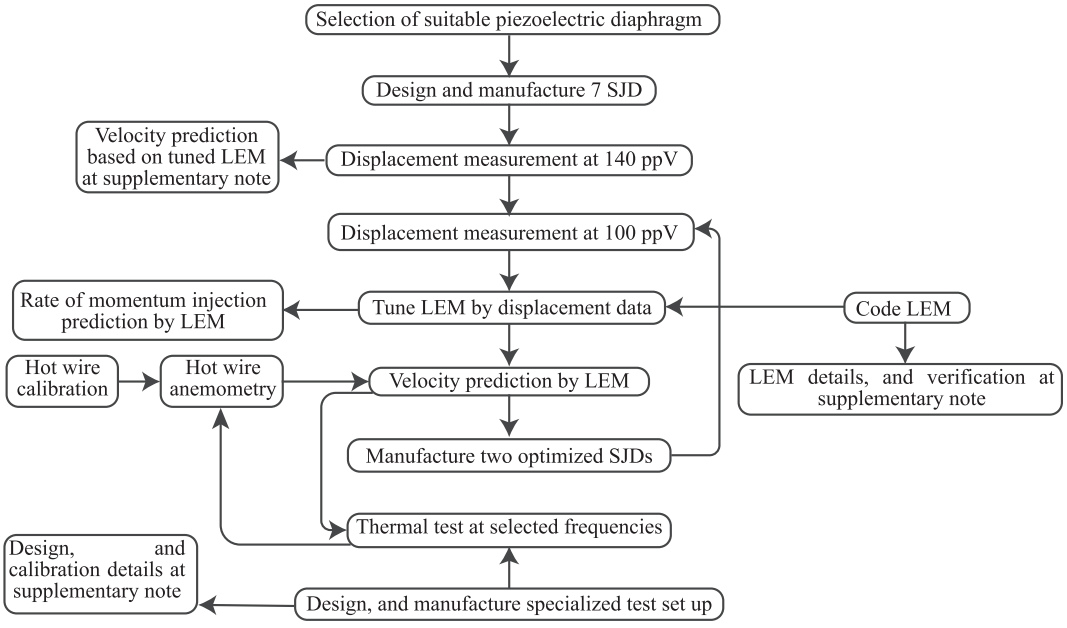


Fig. 10. Research organization flowchart.

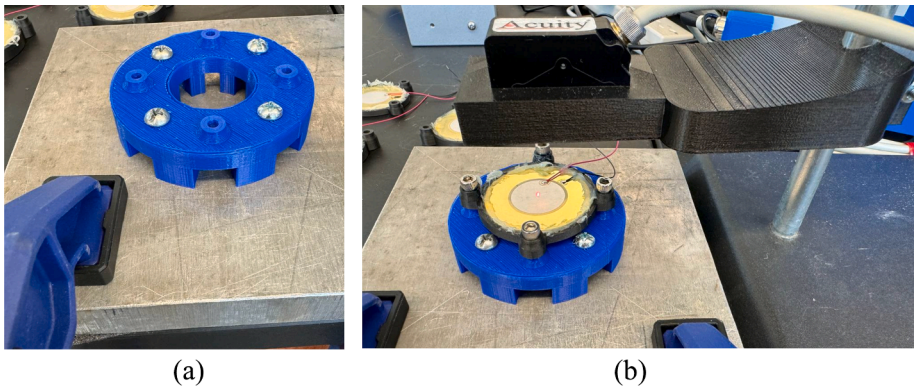


Fig. 11. (a) Test bed for securing SJD and (b) showcasing installation of SJD on test bed and measurement of displacement using laser triangulation sensor.

Siglent SDG 2042X function generator generates sine signals with an output amplitude of 0.7 V, which are amplified by a Trek PZD700 High Voltage Amplifier with a gain range of 200 to 140 V_{pp}. Later, after a few devices (SJD 1, 2, and 7) failed at this voltage, the voltage was reduced to 100 V_{pp}. Details of measurements in 140 V_{pp} voltage are provided in supplementary notes. Displacement measurements were carried out on two separate days. Starting at 25 Hz, the SJDs were operated for 10 s at each frequency increment of 25 Hz up to 800 Hz to capture the actuator's resonance. After being disconnected from the test bed, the SJDs were reinstalled and tested the next day, beginning at 2100 Hz and decreasing in 100 Hz increments. This procedure aimed to investigate the Helmholtz resonance and correlate it with the previous day's results.

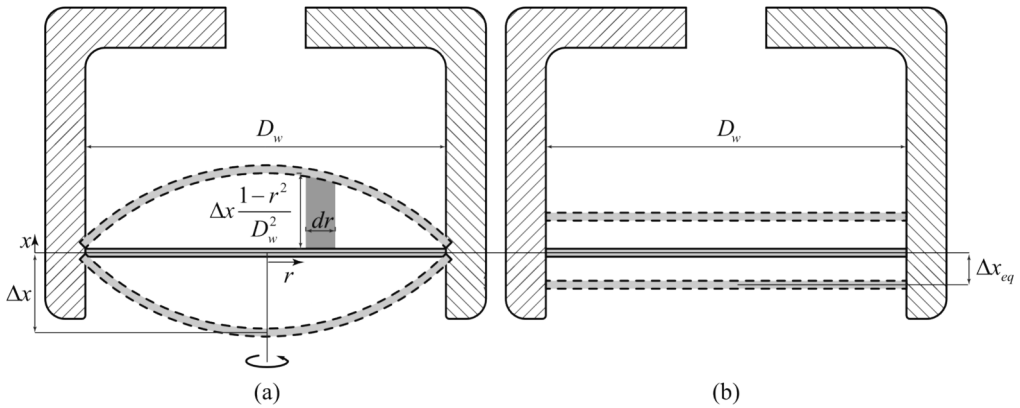


Fig. 12. Deflection behavior of the diaphragm.

It was discovered that each measurement targeted slightly different points on the disc, and that the system's stiffness varied slightly with each installation due to the use of polymer body. Despite these variations, these testing methods were successful in detecting consistent behavior in SJDs.

The resonance of the wall allows for an estimate of the structure's spring constant, but this is only an approximation. This is due to the resonance being observed in a coupled system involving both acoustic mass and stiffness. Assuming that the deflection at the lowest measured frequency (25 Hz) corresponds to the static deflection, the force exerted by the piezoelectric ceramic patch on the diaphragm can be estimated. To match the deflection behavior observed in the SJDs, ill-defined parameters in the LEM code is adjusted, the C_l.

It needs to be noted that the deflection behavior of the diaphragm can be approximated as shown in Fig. 12. This is while in LEM the model assumes one dimensional (piston-in-cylinder) movement of the piezoelectric disc bender. In order to match the behavior of the experiment to the LEM model, one method is to evaluate the equivalent value of Δx for the LEM model Δx_{eq}. As a result, we adjust the volume change in the cavity to match the volume change of cylindrical motion from the LEM model.

The behavior of diaphragm bending can be modeled as the static deflection of a circular membrane clamped on the edge under a uniform load. The deflection profile can be estimated using [57]:

$$\Delta x(r) = \Delta x(1 - 4(r/D_w)^2 + 8C(r/D_w)^2 \ln(2r/D_w)) \tag{17}$$

Where C is 1 for clamped and 0 for pinned case. Most of the time even when the piezoelectric disc is being clamped deflection behavior is towards the pinned condition as seen in [66].

In our case, because the disc is held in place with flexible adhesive, we assume that deflection is completely pinned (adhesive undergoes strain in the edge, keeping disc bending behavior not affected at boundaries). This is further supported by measurement data showing disc's low resonance frequency and high deflection.

Volume change in the experiment can be considered as:

$$\Delta V_0 = \int_0^{D_w/2} \Delta x\left(1 - 4(r/D_w)^2 \cdot (2\pi r)\right) dr = \Delta x\left[\pi r^2 - \frac{2\pi}{D_w^2} r^4\right]_0^{D_w/2} = \frac{\pi}{8} \Delta x D_w^2 \tag{18}$$

While in case of LEM one can write:

$$\Delta V_0 = \frac{\pi}{4} \Delta x_{eq} D_w^2 \tag{19}$$

where:

$$\Delta x_{eq} == 0.5 \Delta x \tag{20}$$

4.2. Synthetic jet impingement experiments

In order to study the impingement cooling performance of designed SJDs, a specialized heater and measurement unit is manufactured. This unit is designed to provide a surface heat flux on a 30 mm × 30 mm area on the top surface of a 30 × 30 × 2.25 mm^3 aluminum block. An array of eleven T type beaded thermocouples (TCs) from Omega (5TC-TT-T-30–36-ROHS) was inserted into the center of the block, just below 0.5 mm from its top surface. Ambient temperature is also measured using T type thermocouple. To provide a heat source to the aluminum block, a self-adhesive polyimide flexible surface heater (model KHLVA-101(1) from Omega) is installed on the bottom of the aluminum block. Power to the heater is supplied by Agilent E3633A DC power supply. Furthermore, the voltage across the heater is monitored by Hewlett 34401A multimeter. Heater control and data acquisition are performed using a custom LabView program. For a robust design, the heater, thermocouples, and the aluminum block were integrated using insulating Polyurethane foam [68]. The thermal resistance of the heater unit is determined using a comprehensive calibration process described in the supplementary note. Fig. 13 presents an overview of the test setup.

The designed heater system is located in the center of a motorized 3D axis's Z-axis. SJDs are mounted on a motorized X-Y axis with the ability to change the angle of impingement. Next, SJDs are aligned with the center of the heater unit and tested at various L_h/D values. Heater power is set to 1.42 W and generates 1578 W/m^2 resulting in an average surface temperature of 74 °C at natural convection at lab temperature of

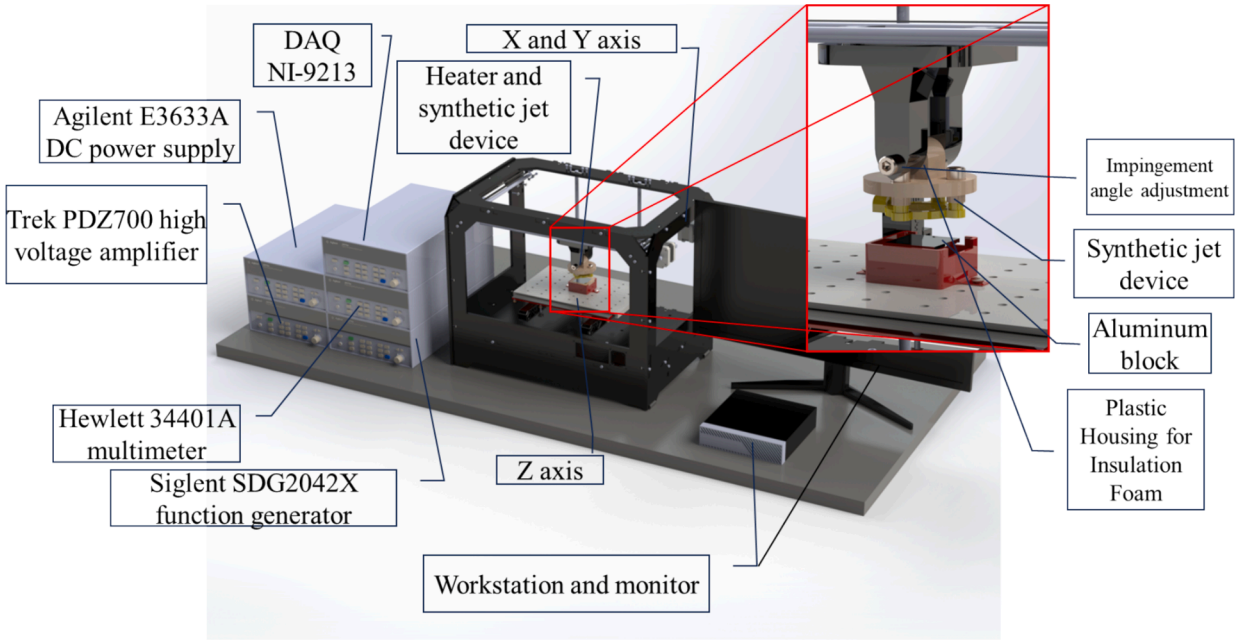


Fig. 13. Test set up preview of SJ impingement heat transfer study.

22–23 °C. The heater elements and heated block are similar in size, preventing heat spreading, thus all eleven thermocouples showed the same temperature with a difference of ± 0.2 K. The heat loss of the heated unit is determined to calculate an accurate heat transfer coefficient from the top surface of the heated block. Thermophysical properties of air are calculated at film temperature using a fourth-order polynomial with temperature dependent properties in atmospheric pressure. After reaching the steady state in natural convection (Nu = 2.3–2.7), SJDs are activated and operated until no temperature variation or steady state in each condition is achieved. SJDs are first investigated at the furthest distance to the heater, from L_h/D of 9 to 3. Larger values were not possible due to the size limitations of the 3D axis.

4.3. Jet exit velocity measurements with a hot wire anemometer

Maximum jet exit velocity measurements (U) were conducted using a DANTEC MiniCTA System equipped with a miniature hot wire probe, a two-point hot-wire calibrator, and StreamWare software. The probe was connected to the MiniCTA module, following the manufacturer's specifications for setup and configuration. Calibration was performed in a controlled environment using the two-point calibrator provided by DANTEC. The probe was exposed to two precisely known velocities, and the corresponding voltage outputs were recorded to extract a 4th degree polynomial calibration function [69]. During experiments, the hotwire probe was positioned at the orifice exit using Velmex precision motion-control 2D axis ensuring alignment perpendicular to the flow direction to minimize directional bias (see Fig. 14). The MiniCTA system was set to sample at kHz levels depending on operation frequency, and data was collected over a 10-second duration for each frequency (starting at 100 Hz by 100 Hz increment). StreamWare software facilitated real-time data acquisition and monitoring. To assess the reliability of the manufacturing process, hotwire measurements were repeated after a 4-month period to evaluate the performance and stability of both the device and the adhesive used in its assembly.

4.4. Power consumption measurement

The applied voltage and current consumption of devices during operation were measured by connecting the Trek PZD700 analog input to the digital voltage converter NI–9205. Measurements were done at 4 kHz for 250 ms during experiments. 1/200 voltage monitor and 0.1 V/mA voltage output were used to evaluate the current and voltage characteristics of SJDs. The power consumption rate calculation was done by integrating the absolute value of the product of voltage and current over a second of operation in MATLAB.

5. Results and discussion

5.1. Deflection measurement, LEM tuning, and hotwire results at 100 V_{pp}

Table 4 presents a summary of the deflection measurement results and LEM parameter tunning at 100 V_{pp}. It needs to be stated that results are presented for SJD-3, 4, 5, and 6 only. Results for all devices in 140 V_{pp} are presented in supplementary notes.

Based on the deflections at 25 Hz one can estimate the force being applied to the diaphragm by the piezoceramic. The applied force is estimated by tuning the LEM with equivalent deflection (Δx_{eq}) behavior from experiments. It can be seen that the maximum force at 100 V_{pp} is in the range of 1.8—2.5 N. At 25 Hz, SJD-4 exhibits the maximum deflection, demonstrating the greatest compliance among other SJDs. The observed differences in compliance and deflection are attributed to several factors: added stiffness from the air inside the cavity, variations in diaphragm properties including location and amount of solder, and inconsistencies in adhesive application. It can also be seen that these factors lead to varying wall resonance measurements for each SJD. Specifically, SJD-4, which has the smallest cavity volume, exhibits the lowest resonance frequency, while SJD-6, with the larger cavity volume, shows resonance at 675 Hz. This issue can be explained by looking at Fig. 4(a). In SJD-4, the deflection of the diaphragm is reduced by applying cavity due to added acoustic mass to the system, on the other hand in SJD-5 and 6 wall resonance increases from the nominal natural frequency of the diaphragm due to added acoustic stiffness to the system. The largest deflection at resonance, 289.3 μm, is observed in SJD-3, which has the largest orifice among other SJDs suggesting that tightening the orifice can reduce deflection. Furthermore, it can be seen that C_l is related to the length of the SJDs, with longer SJDs resulting in lower C_l values. It is important to note that C_d, the nozzle discharge coefficient, is kept constant at 0.6, which is an accurate assumption. According to a classic study by [70], C_d in small β values is almost constant in the variety of Re ranges, thus it can be kept constant throughout all stages of SJ generation. However, future research can focus on C_l's temporal behavior, which will improve the accuracy of LEM in predicting SJ behavior.

Apart from wall resonance, the Helmholtz resonance of the SJDs can be estimated by observing the slight increases in deflection at specific frequencies, with resonances occurring between 1500 and 1800 Hz. Notably, increasing the size of the orifice lowers the Helmholtz resonance frequency (see Eq. (15). An intriguing pattern is observed with

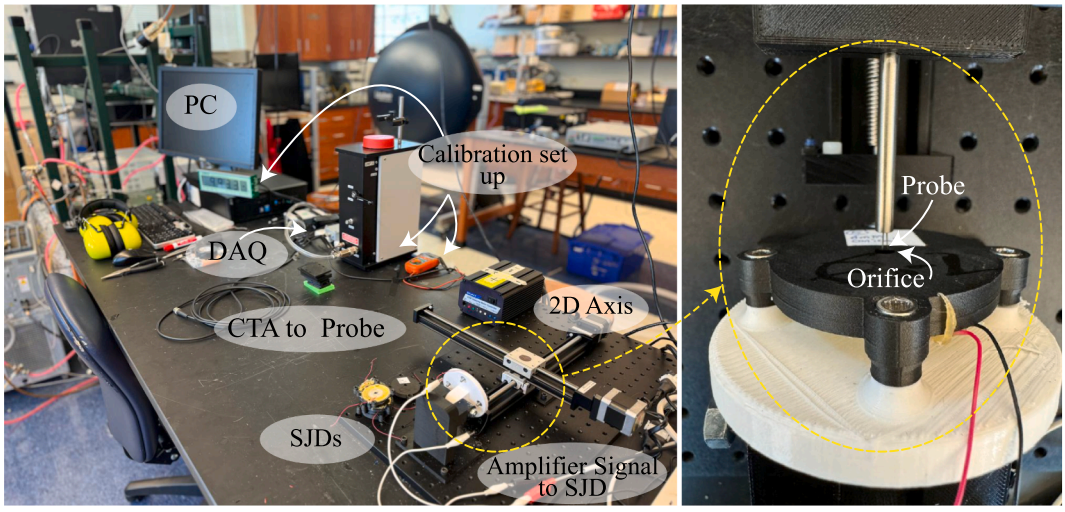


Fig. 14. Hotwire anemometry set up overview.

Table 4
Results of deflection test and LEM tuning.

		Deflection Experiment						*LEM*				
SJD #		fr_{min} (Hz)	Δx (μm)	fr_w (Hz)	Δx (μm)	fr_H (Hz) Experimental	Δx (μm)	$k \times 10^4$ (N/m)	F_{max} (N)	C_l	C_d	fr_H (Hz) LEM
3	con	25	62	600	289.3	1600–2000	54	8.49	1.8	2	0.6	1370
4	con	25	100	500	245	1500–1800	61	7.49	2.11	2.75	0.6	1618
5	con	25	79	625	231	1600–1900	37	7.65	2.49	0.45	0.6	1556
6	con	25	58	675	193	1500–1900	23	9.58	2.41	0.3	0.6	1600

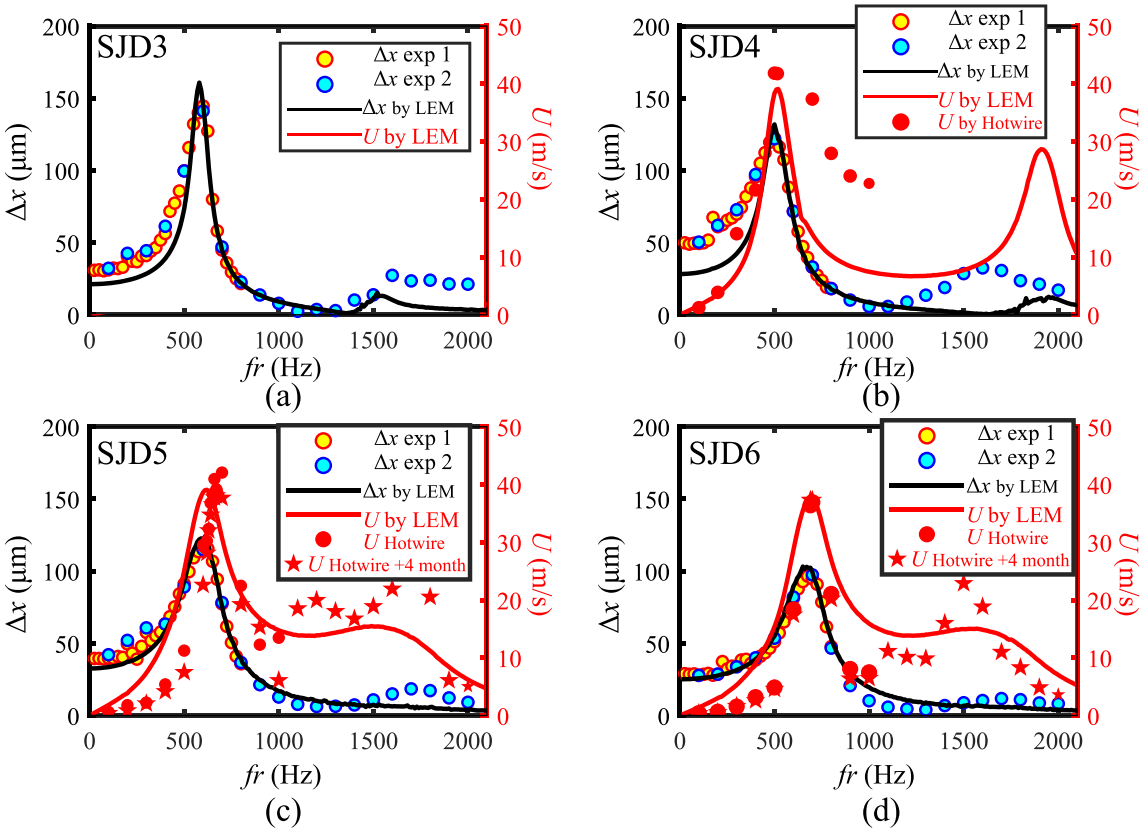


Fig. 15. Displacement behavior of the SJD (a)3, (b)4, (c)5, and (d)6 in two different measurements vs the deflection behavior provided by tuned LEMs. Maximum area averaged velocity performance of the SJDs based on LEM and hotwire is also provided on the right axis.

respect to cavity thickness: thinner cavities (smaller L – see Fig. 3(a)) exhibit more pronounced effects on wall deflection in Helmholtz resonance. For instance, in a cavity with a length of 1 mm, the largest deflection in Helmholtz resonance measured is 91 μm. When the diaphragm is closer to the orifice, oscillating air in and near the cavity can be reinforced by the diaphragm's oscillation, resulting in oscillation of the wall and vice versa. As shown in SJD 5 and 6, increasing L reduces this effect.

Fig. 15 presents the frequency-dependent displacement measurements, with "exp 1" in legend corresponding to day 1's measurements starting at 25 Hz and increasing in increments of 25 Hz up to 800 Hz, and "exp 2" representing day 2's measurements beginning at 2100 Hz and decreasing in 100 Hz increments down to 100 Hz. The results from the tuned LEMs are depicted with solid lines for each scenario. A good behavior matching between trends of Δx by LEM and measurements can be seen. However, for SJD-3 and 4, which have a shorter cavity length (L), the intensification of wall deflection near the Helmholtz effect is not captured well enough with the LEM model. The main reason for this could be due to the fact that these SJDs have conical cavity shapes, and this phenomenon is ignored in LEM models for these SJDs (the cavity is considered cylindrical in LEM). However, for SJD-3 and 4, the model predicted the results well without requiring a change in the L value. Nonetheless, the LEM model considers one-dimensional velocity and needs specific tuning to capture real-time device behavior. Table 4 also includes the evaluated Helmholtz resonances by LEM models.

The peak instantaneous area averaged jet exit velocity (U) predicted by the LEM model for each SJD is also provided on Fig. 15, shown by a red line with an axis on the right side of the graphs. It can be seen that maximum U occurs with a slight offset from resonance frequency due to phase lag. For SJD 3 maximum U is evaluated at 585 HZ with the value of 29.6 m/s. For SJ 4 to 6, these values are 515 Hz with 39.1 m/s, 620 Hz with 39.1 m/s, and 690 Hz with 37.8 m/s. These values are summarized in Table 5. It can be seen that SJD 3, and 4 show high jet exit velocity at Helmholtz resonance, in contrast to SJD 5, and 6.

Measured velocity (U) values by hotwire probe are also shown in

Table 5
Predicted maximum area averaged jet exit velocity.

SJD #	fr (Hz)	U (m/s)	U_{hw} (m/s)
3	585	29.6	–
4	515	39.1	41.8
5	620	39.1	30.42
6	690	37.8	36.4

Fig. 15, represented by red circles with an axis on the right side of the graphs (SJD3 failed due to accidental overload). Hotwire measurements are evaluated using a 100 Hz increment to determine U near wall resonance. For SJD 4, the maximum U is measured at 600 HZ and is 50.5 m/s. For SJ 5 and 6, the values are 700 Hz with 42.1 m/s and 700 Hz with 36.9 m/s. It is clear that LEM has a slight underestimation of the frequency where the maximum U is achieved, as well as the value of U itself. Probable causes can be stiffness change of the system after each installation of SJDs in the measurement stage, simplification of the deflection profile shown in Fig. 12, and intrinsic simplification of flow behavior by LEM itself. Tuned LEM shows overall acceptable agreement in predicting U and its frequency response, while only deflection measurements are used to tune it. In Fig. 15(b) for SJD 4 with the thinnest cavity profile, at above wall resonance frequencies, a sharp drop in U values predicted by LEM cannot be seen in experimental data. This suggests that displacement after wall resonance is more pronounced in thinner profiles, and any SJDs designed to operate at Helmholtz resonance (rather than wall resonance) can benefit from shorter cavity lengths. Thus, SJDs with thinner cavities perform better in Helmholtz resonances. The benefit of this is that the diaphragm will require less input power and deflection, allowing the SJD to operate more efficiently and reliably. Table 5 provides a summary of maximum U prediction by LEM and compares them by hotwire measurements.

Hotwire tests were conducted on SJA 5 and 6 after a 4-month period to evaluate the reliability and stability of the manufacturing process. These tests were performed at frequencies up to 2000 Hz. The results demonstrate a close match with the initial measurements, indicating that the SJAs maintained their performance after 4 months in a lab environment. In higher frequencies, the LEM model, tuned based on deflection data, underestimates the velocity. Despite this, the model accurately predicts the Helmholtz resonance frequency, as shown by the alignment of the second velocity peak. Additionally, the model accurately predicts the velocity drop beyond the Helmholtz resonance range, particularly near 2000 Hz.

5.2. Deflection measurement, LEM tuning, and hotwire results at 100 V_{pp} for high-velocity ARTgroup SJDs

Table 6 summarizes the deflection measurement results at 100 V_{pp} for optimized SJDs. The results indicate that with smaller orifice, systems become stiffer that shift wall resonance to higher frequencies (see Fig. 4(a)). Conversely, increasing the length of the orifice (L_o) and reducing its size seems to successfully align Helmholtz resonance with wall resonance.

With the same geometrical characteristics, it can be seen that the -wall resonance of the conical cavity is higher than cylindrical cavity. Thus, as it can be seen in Table 6, the conical system's k is nearly twice that of the cylindrical cavity. Compared to previous SJDs, it is important to note that acoustic effects become more pronounced as the orifice becomes smaller. Therefore, in the LEM for SJD8-con, the length of the cavity is multiplied by a factor of 0.58 to match experimental results.

Fig. 16 presents the frequency-dependent displacement measurements for optimized SJDs. The results from the tuned LEMs are depicted with solid lines for each scenario.

Regarding diaphragm deflection, good matching can be observed with LEM and experimental results. It can be seen that since both resonance modes overlap each other maximum velocity output and maximum deflection output occur in the same resonance modes where maximum velocity at 840 Hz of 94 m/s is theoretically evaluated for SJD8-cyl and 94.48 m/s at 1130 Hz for SJD8-con. These values are summarized in Table 7. Similar to previous cases, LEM underestimates the actual jet exit behavior velocity. The maximum velocities recorded by hotwire measurements were 102.3 m/s at 900 Hz for SJD8-cyl and 106.1 m/s at 1130 Hz for SJD8-con. It should be noted that experimental data show a second peak in U for SJD8-cyl at 1300 Hz with 72.6 m/s and at 1400 Hz with 105.7 m/s. This is more highlighted in second hotwire data. This might suggest that LEM's optimization (or matching of Helmholtz and wall resonance) was unsuccessful. However, when we examined wall deflection behavior, we could not detect a rise in deflection behavior at these frequencies. This increase in jet exit behavior might be due to higher order acoustic effects in SJDs at these frequencies. The results also show that the manufacturing process is capable of achieving the highest reposted SJ velocities using a single piezoelectric diaphragm at actuation frequencies below 1 kHz.

5.3. Heat transfer test results

Fig. 17 provides the enhancement of natural convection (Nu/Nu_{nc}) with the presence of SJDs at different distances between the orifice and the heater, L_h/D. Furthermore, it shows the achieved heat transfer coefficient at room temperature. Due to the larger size of the orifice in SJD 3, L_h/D of 2 to 6 is considered for this device, while the rest of the SJDs are tested in L_h/D of 3 to 9. Devices are tested in frequencies shown in Tables 5 and 7.

The mutual behavior of SJDs is that the heat transfer coefficient increases with increasing distance. As the SJD approached the surface, air began to mix in large vortical structures trapped between the heated wall and the SJD body, increasing film temperature and reducing SJ's cooling capacity. SJD 3, 5, and 6 exhibited similar heat removal behavior. SJD 4 exhibited similar behavior, but it had a slightly lower heat transfer coefficient at L_h/D of 7 and 8. It is clear that optimized devices, despite having higher jet exit velocities, did not provide as much heat transfer capability as SJDs with larger orifice sizes. This emphasizes the importance of equations (9) and (10) in SJD behavior analysis. Despite having a 2 to 3 times higher jet exit velocity, SJs generated by SJD 8-cyl and 8-con are thermally inferior due to their lower impulse injection rate (I).

To understand this, the rate of injected momentum towards the heater I (see Eq. (10)) is evaluated based on the LEM model and is provided in Fig. 18(a). There is a clear correlation between I in Fig. 18 (a), with the achieved thermal performance shown in Fig. 17. In other words, by following the peaks of I, it can be observed that the higher the generated momentum rate, the better the device performs, even though the jet can have lower jet velocity. This correlation is particularly significant considering that the surface temperature of the heater in this experiment can be considered uniform due to the nature of its design. Thus, SJDs with small orifices can produce high heat transfer coefficients from a local perspective, but their overall (useful) heat removal behavior can be related to I. This aspect is occasionally overlooked when tuning SJDs for specific tasks. In other words, optimizing piezoelectric based SJDs solely to achieve higher jet exit velocity may be misleading, and optimization should also consider I.

Fig. 18(b) shows the flow rate ejected towards the heated surface, with an interesting contrast between I and flow rate trends.

Table 6
Results of deflection test and LEM tuning for optimized SJD-8.

	Deflection Test						*LEM*				
SJD #	fr_{min} (Hz)	Δx (μm)	fr_w (Hz)	Δx (μm)	fr_H (Hz) Experimental	Δx (μm)	$k \times 10^4$ (N/m)	F_{max} (N)	C_l	C_d	fr_H (Hz) LEM
8-cyl	25	94	850	265	–	–	6.52	2.58	1.5	0.6	–
8-con	25	77.21	1175	128	–	–	12.2	3.05	1.5	0.6	–

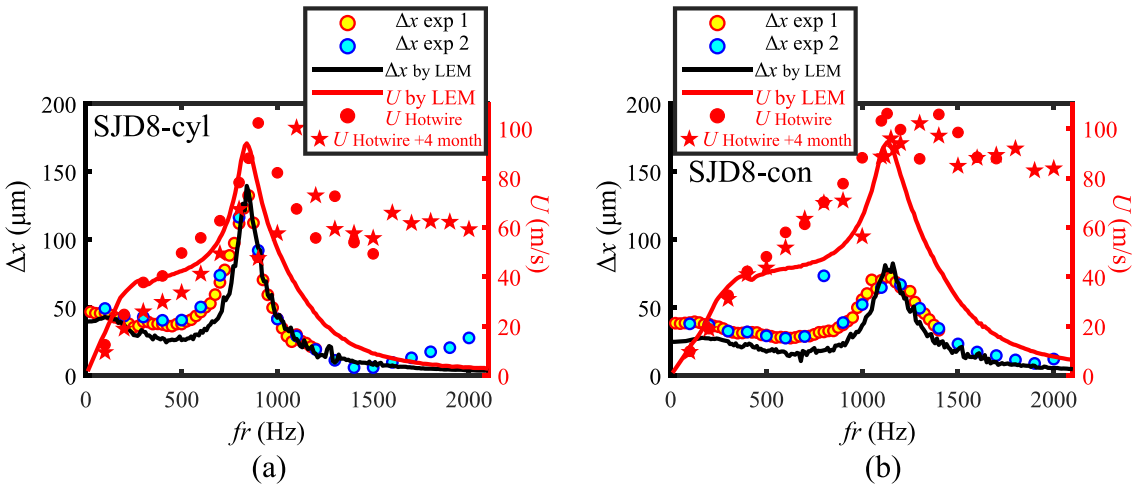


Fig. 16. Displacement behavior of SJD (a)8-cyl, and (b)8-con in two different measurements vs the deflection behavior provided by tuned LEMs. Maximum area averaged velocity performance of the SJDs based on LEM is also provided on the right axis.

Table 7
Predicted maximum U for optimized SJDs.

SJD #	*fr* (Hz)	*U* (m/s)	U_{hw} (m/s)
8-cyl	840	94.00	87.2
8-con	1130	94.48	106.1

Furthermore, it is interesting to see that SJs, by a fraction of flow rate in comparison to fan systems, are capable of similar cooling performance, highlighting their superior impingement characteristics in comparison to dispersed air impingement. For example, the maximum average heat transfer coefficient observed here (120 W/m^2-K) is comparable to the local heat transfer coefficient achieved by a 59 mm diameter axial fan at 2000 rpm [71], which generally consumes about 1.5 W of power at this speed. Similarly, the average heat transfer coefficient (120 W/m^2-K) achieved by high-speed axial fans operating at 9000 rpm [72], with a typical power consumption rating of 3–5 W at this rpm, is matched by our SJDs. Power consumption figures for SJD 5 at 670 Hz were evaluated at 0.66 W, for SJD 7 at 700 Hz at 0.49 W, for SJD 8-cyl at 1100 Hz at 1.12 W, and for SJD 8-con at 1.20 W.

Table 8 provides a summary of performed thermal tests. It can be observed that even when operated at different frequencies, SJDs with similar *I* performance show similar Nu number values. Therefore, from an average heat removal capability perspective, we can conclude that the performance of SJDs is more correlated to the distance of impingement and their capability of momentum generation rate Nu ≈ Nu(L_h/D, *I*).

6. Conclusion

This study presents a novel 3D printing method for fabricating SJDs that offers significant advantages for researchers. This method enables rapid prototyping, complete design freedom, and the creation of ultra-thin devices (down to 4 mm) without requiring mechanical fasteners. The fabricated SJDs achieved high jet exit velocities of up to 106 m/s using a single piezoelectric diaphragm, which is among the highest reported values. The deflection profile of the diaphragm in these devices closely resembles a pinned disc boundary condition, which enables high deflection rate. 3D-printed SJDs demonstrated reliable operation even at high voltage levels (100 V_{pp}) over several months of testing under various heated and installation conditions. The primary failure mode observed was cracking of the piezoceramic patch in piezoelectric diaphragms when the central deflection exceeded 400 μm (in 44 mm shim diameter).

LEM proved to be a reliable approach for characterizing SJDs with orifices located on the axis of rotation. This study demonstrates that tuning the LEM based solely on diaphragm deflection behavior is sufficient to predict the fluidic behavior of manufactured SJDs with reasonable accuracy. However, a slight underestimation of jet exit velocity was observed at Helmholtz resonance frequencies, likely due to the model or measurements not fully capturing wall behavior at these frequencies or neglecting other acoustic/vibration effects. Despite this limitation, LEM with deflection-based tuning remains a valuable tool for researchers designing SJDs, particularly when hotwire anemometry measurements are not feasible.

This research also highlights the importance of hydrodynamic impulse generation rate over jet exit velocity alone when optimizing SJDs for heat removal applications. Devices with lower jet exit velocities but

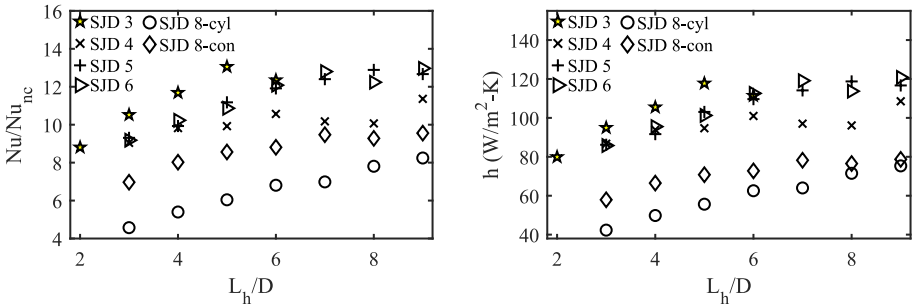


Fig. 17. Factor of Nu number enhancement in comparison to steady state natural convection case (left), and heat transfer coefficient (right) in different orifice to heat place distance ratios.

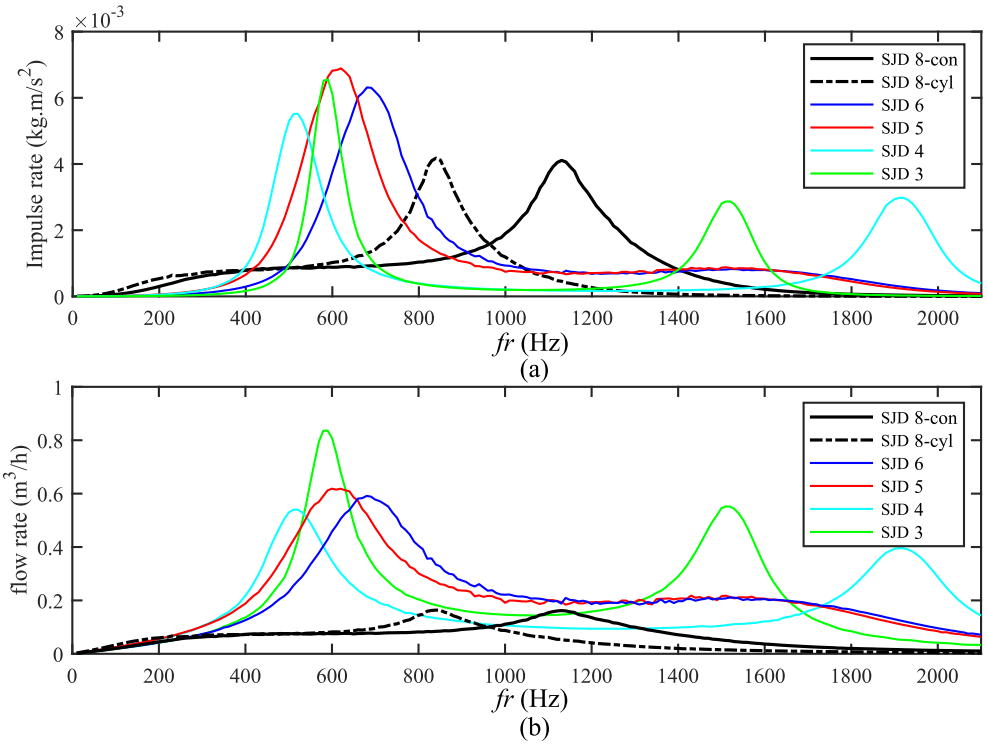


Fig. 18. (a) Rate of injected momentum (I), and (b) rate of injected flow in different operation frequencies evaluated by LEM.

Table 8
Summary of thermal test results.

SJD #	*Voltage* (V_{pp})	*fr* (Hz)	Nu_{nc}	Nu							
				$L_h/D = 9$	$L_h/D = 8$	$L_h/D = 7$	$L_h/D = 6$	$L_h/D = 5$	$L_h/D = 4$	$L_h/D = 3$	$L_h/D = 2$
3	100	585	2.6	–	–	–	31.5	33.3	29.8	26.8	22.4
4	100	515	2.7	30.6	27.1	27.4	28.5	26.8	26.5	24.4	–
5	100	620	2.6	33.0	33.6	32.3	31.0	29.1	25.9	24.2	–
6	100	690	2.6	34.1	32.2	33.7	31.8	28.6	26.9	24.1	–
8-cyl	100	840	2.6	21.2	20.1	18.0	17.5	15.5	13.9	11.8	–
8-con	100	1130	2.3	22.1	21.5	21.9	20.4	19.8	18.6	16.1	–

higher impulse rates achieved superior heat transfer through synthetic jet impingement. This finding suggests that a focus on impulse generation, which incorporates both velocity and mass flow rate, provides a more comprehensive parameter for optimizing SJD heat transfer performance.

Finally, the study confirms previous observations regarding the existence of an optimal spacing between the SJD and the target surface for maximizing heat transfer. Orifice distance to heated plated ratio to orifice diameter of 8 or above is recommended. In this distance and above SJDs achieved a significant (over 13 times at 120 W/m^2-K) enhancement in heat transfer compared to natural convection, using only a fraction of the mass flow rate typically required by conventional cooling methods (less than 1 m^3/hr). Additionally, using less than 1.12 W, the average heat transfer coefficients achieved by these devices rival those of high-speed axial and centrifugal fans at 3-5 W power consumption.

CRediT authorship contribution statement

Mohammad Azarifar: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Faisal Ahmed**: Investigation. **Mehmet Arik**: Supervision, Funding acquisition, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.applthermaleng.2024.124914.

Data availability

Data will be made available on request.

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